10. Closest Pair of Points – Brute Force

Aim

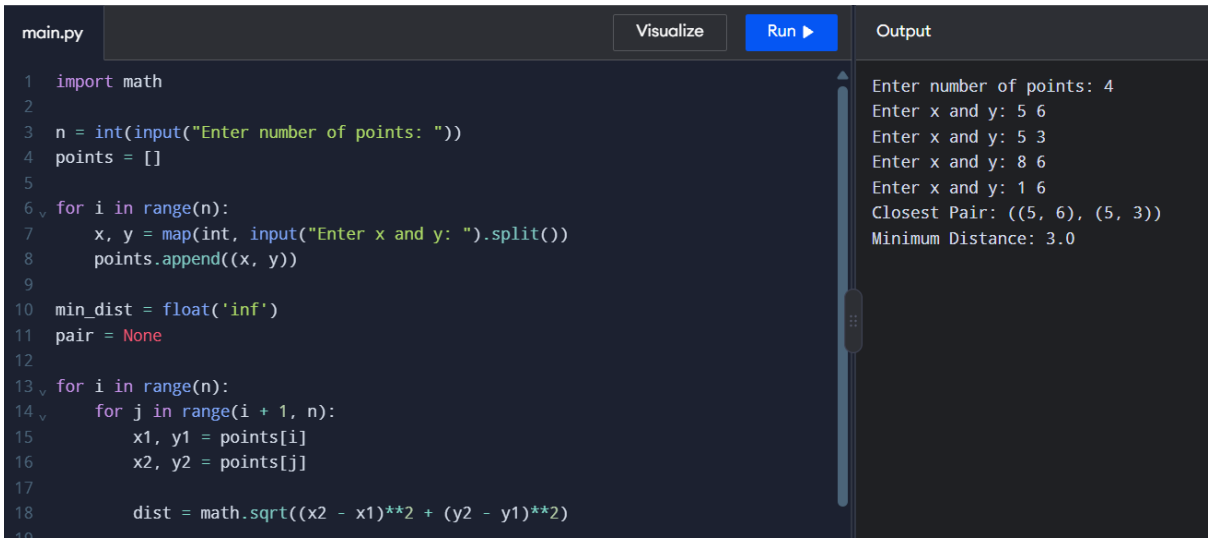
To find the closest pair of points in a two-dimensional plane using the Brute Force algorithm and analyze its time complexity.

Algorithm

1. Read the coordinates of \(n\) points.
2. Initialize the minimum distance to infinity.
3. Compare every pair of points.
4. Calculate the Euclidean distance between each pair.
5. Update the minimum distance and closest pair.
6. Display the closest pair and its distance.

Procedure

1. Start the program.
2. Read the number of points and their coordinates.
3. Use nested loops to select every possible pair.
4. Calculate the distance using the Euclidean distance formula.
5. Update the closest pair whenever a smaller distance is found.
6. Display the closest points and minimum distance.



Result

Thus, the closest pair of points was successfully identified using the Brute Force algorithm, and its time and space complexity were analyzed.